Spring potential energy

antocciphysics | TEACHER KEY | CPA

Learning target: Calculate stored spring energy and distinguish it from force.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 12; 3 course days later.*
15 J, by conservation of mechanical energy.

02. Explain a worked example (worked)
Both triangle height and base double, so area and stored energy quadruple.

03. Look and explain (model)
Energy 4 times; force 2 times.

04. Build the solution (practice)
x^2=.04 m^2; U 2 J.

05. Use the idea (reason)
x^2 positive; force has direction, energy scalar measured relative equilibrium.

06. Choose a method (mixed) *Review: Unit 4, Day 3; 12 course days later.*
-24 J; the force opposes displacement.

07. Exit check (exit)
12 J within Hooke’s law range.

08. STRETCH: Apply it another way (stretch)
0.25, 1, 2.25 J; curved quadratic increase.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 12; 3 course days later.*
No. That height requires an increase mg×2 in potential energy, exceeding the initial energy; it would require negative final K.

02. Explain a worked example (worked)
Both triangle height and base double, so area and stored energy quadruple.

03. Represent and interpret (model)
Energy 4 times; force 2 times.

04. Solve and justify (practice)
U 1 J; $v=\sqrt{(2/.5)}=2$ m/s.

05. Analyze the reasoning (reason)
x^2 positive; force has direction, energy scalar measured relative equilibrium.

06. Choose a method (mixed) *Review: Unit 4, Day 3; 12 course days later.*
W=-24 J because cos 120°=-1/2; the component opposes displacement.

07. Exit check (exit)
$|x|=\sqrt{(.8/80)}=.10$ m; force 8 N.

08. STRETCH: Apply it another way (stretch)
Energy 4.5 J; $v=\sqrt{18}\approx4.24$ m/s. Compression 0.30 m doubles speed because spring energy is quadratic.